

Task 2

Implement the numerical solution using finite difference method for the following partial differential equation (no need to compare with any analytical solution):

$$\frac{\partial c}{\partial t} = D \frac{\partial^2 c}{\partial x^2}, \quad 0 \leq x \leq 1$$

Boundary conditions:

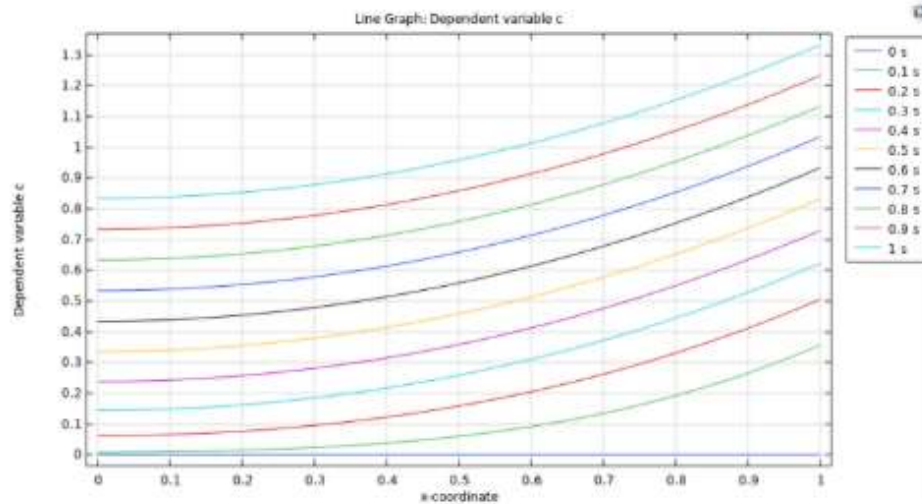
$$\text{At } x = 0, \quad \frac{\partial c}{\partial x} = 0,$$

$$\text{At } x = 1, \quad \frac{\partial c}{\partial x} = 1$$

Initial condition:

$$\text{At } t = 0, \quad c = 0.$$

With $D = 1$, your solution should come out as follows:



Analytical solution:

Task 2

$$\frac{\partial c}{\partial t} = D \frac{\partial^2 c}{\partial x^2}$$

$$\frac{\partial c}{\partial t} \approx \frac{c_i^{n+1} - c_i^n}{\Delta t} \quad \left. \vphantom{\frac{\partial c}{\partial t}} \right\} \text{time derivatives}$$

$$\frac{\partial^2 c}{\partial x^2} \approx \frac{c_{i+1}^n - 2c_i^n + c_{i-1}^n}{\Delta x^2} \quad \left. \vphantom{\frac{\partial^2 c}{\partial x^2}} \right\} \text{space derivatives}$$

$$\therefore \frac{c_i^{n+1} - c_i^n}{\Delta t} = D \frac{c_{i+1}^n - 2c_i^n + c_{i-1}^n}{\Delta x^2}$$

$$c_i^{n+1} = c_i^n + \underbrace{\frac{D \Delta t}{\Delta x^2}}_{r \text{ (let)}} (c_{i+1}^n - 2c_i^n + c_{i-1}^n)$$

$$\text{at } t=0 \quad c(x, 0) = 0$$

$$\& \text{ at } x=0 \quad \frac{\partial c}{\partial x} = 0$$

$$\Rightarrow \frac{c_2 - c_1}{\Delta x} = 0$$

$$\Rightarrow c = \text{const} \quad (c_1 = c_2)$$

$$\text{at } x=1 \quad \frac{\partial c}{\partial x} = 1$$

$$\frac{c_N - c_{N-1}}{\Delta x} = 1$$

$$c_N = c_{N-1} + \Delta x$$

Matlab Implementation:

```
clc;
clear;
close all;

D = 1;

Nx = 21;
dx = 1/(Nx-1);

dt = 0.001;

T = 1;

Nt = T/dt;

r = D*dt/dx^2;

x = linspace(0,1,Nx);

c = zeros(Nx,1);

figure;
hold on;

for n = 1:Nt

    c_old = c;

    for i = 2:Nx-1

        c(i)=c_old(i)+r*( ...
            c_old(i+1)-2*c_old(i)+c_old(i-1));

    end

end
```

```
c(1)=c(2);  
c(end)=c(end-1)+dx;
```

```
t=n*dt;
```

```
if abs(mod(t,0.1))<dt
```

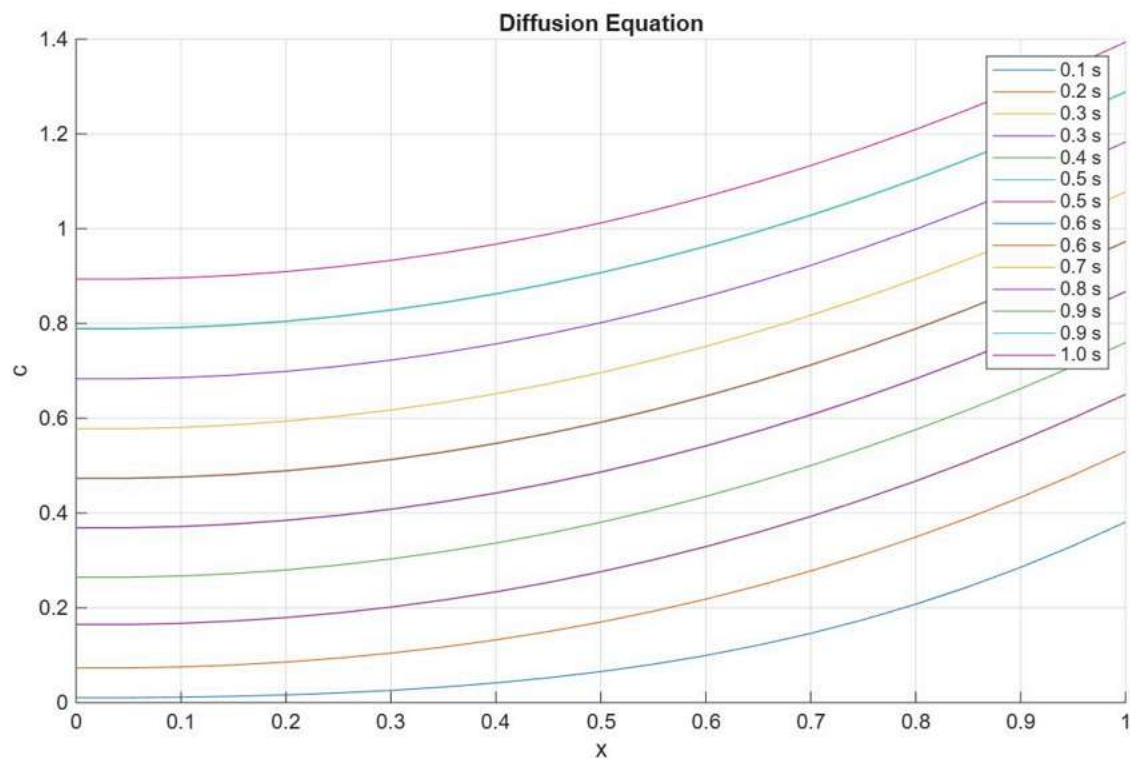
```
    plot(x,c,'DisplayName',sprintf('%.1f s',t));
```

```
end
```

```
end
```

```
xlabel('x');  
ylabel('c');  
title('Diffusion Equation');  
legend;  
grid on;
```

Plot obtained:



Github repository: <https://github.com/mugdhajha/Project-Research.git>